History of IBM

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1910s–1950s

IBM originated with several technological innovations developed and commercialized in the late 19th century. Julius E. Pitrap patented the computing scale in 1885;[18] Alexander Dey invented the dial recorder (1888);[19] Herman Hollerith patented the Electric Tabulating Machine (1889);[20] and Willard Bundy invented a time clock to record workers' arrival and departure times on a paper tape (1889).[21] On June 16, 1911, their four companies were amalgamated in New York State by Charles Ranlett Flint forming a fifth company, the Computing-Tabulating-Recording Company (CTR) based in Endicott, New York.[1][22] The five companies had 1,300 employees and offices and plants in Endicott and Binghamton, New York; Dayton, Ohio; Detroit, Michigan; Washington, D.C.; and Toronto, Canada.[23]

Collectively, the companies manufactured a wide array of machinery for sale and lease, ranging from commercial scales and industrial time recorders, meat and cheese slicers, to tabulators and punched cards. Thomas J. Watson, Sr., fired from the National Cash Register Company (NCR) by John Henry Patterson, called on Flint and, in 1914, was offered a position at CTR.[24] Watson joined CTR as general manager and then, 11 months later, was made President when antitrust cases relating to his time at NCR were resolved.[25] Having learned Patterson's pioneering business practices, Watson proceeded to put the stamp of NCR onto CTR's companies.[24]:105 He implemented sales conventions, "generous sales incentives, a focus on customer service, an insistence on well-groomed, dark-suited salesmen and had an evangelical fervor for instilling company pride and loyalty in every worker".[26][27] His favorite slogan, "THINK", became a mantra for each company's employees.[26] During Watson's first four years, revenues reached $9 million ($163 million today) and the company's operations expanded to Europe, South America, Asia and Australia.[26] Watson never liked the clumsy hyphenated name "Computing-Tabulating-Recording Company" and chose to replace it with the more expansive title "International Business Machines" which had previously been used as the name of CTR's Canadian Division;[28] the name was changed on February 14, 1924.[29] By 1933, most of the subsidiaries had been merged into one company, IBM.[30]

NACA researchers using an IBM type 704 electronic data processing machine in 1957

The Nazis made extensive use of Hollerith punch card and alphabetical accounting equipment and IBM's majority-owned German subsidiary, Deutsche Hollerith Maschinen GmbH (Dehomag), supplied this equipment from the early 1930s. This equipment was critical to Nazi efforts to categorize citizens of both Germany and other nations that fell under Nazi control through ongoing censuses. These census data were used to facilitate the round-up of Jews and other targeted groups, and to catalog their movements through the machinery of the Holocaust, including internment in the concentration camps.[31] Edwin Black contends that IBM's dealings with Nazis through its New York City headquarters persisted during World War II.[32] Nazi concentration camps operated a Hollerith department called Hollerith Abteilung, which had IBM machines, including calculating and sorting machines.[33]

IBM as a military contractor produced 6% of the M1 Carbine rifles used in World War II, about 346,500 of them, between August 1943 and May 1944. IBM built the Automatic Sequence Controlled Calculator, an electromechanical computer, during World War II. It offered its first commercial stored-program computer, the vacuum tube based IBM 701, in 1952. The IBM 305 RAMAC introduced the hard disk drive in 1956. The company switched to transistorized designs with the 7000 and 1400 series, beginning in 1958. In which, IBM considered the 1400 series the "model T" of computing, because it was the first computer with over ten thousand unit sales by IBM.[*citation needed*]

In 1956, the company demonstrated the first practical example of artificial intelligence when Arthur L. Samuel of IBM's Poughkeepsie, New York, laboratory programmed an IBM 704 not merely to play checkers but "learn" from its own experience. In 1957, the FORTRAN scientific programming language was developed.[*citation needed*]

1960s–1980s

In 1961, IBM developed the SABRE reservation system for American Airlines and introduced the highly successful Selectric typewriter. Also in 1961 IBM used the IBM 7094 to generate the first song sung completely by a computer using synthesizers. The song was Daisy Bell (Bicycle Built for Two).

In 1963, IBM employees and computers helped NASA track the orbital flights of the Mercury astronauts. A year later, it moved its corporate headquarters from New York City to Armonk, New York. The latter half of the 1960s saw IBM continue its support of space exploration, participating in the 1965 Gemini flights, 1966 Saturn flights, and 1969 lunar mission. IBM also developed and manufactured the Saturn V's Instrument Unit and Apollo spacecraft guidance computers.

An IBM System/360 in use at the University of Michigan c. 1969

IBM guidance computer hardware for the Saturn V Instrument Unit

On April 7, 1964, IBM launched the first computer system family, the IBM System/360. It spanned the complete range of commercial and scientific applications from large to small, allowing companies for the first time to upgrade to models with greater computing

capability without having to rewrite their applications. It was followed by the IBM System/370 in 1970. Together the 360 and 370 made the IBM mainframe the dominant mainframe computer and the dominant computing platform in the industry throughout this period and into the early 1980s. They and the operating systems that ran on them such as OS/VS1 and MVS, and the middleware built on top of those such as the CICS transaction processing monitor, had a near-monopoly-level market share and became the thing IBM was most known for during this period.[34]

In 1969, the United States of America alleged that IBM violated the Sherman Antitrust Act by monopolizing or attempting to monopolize the general-purpose electronic digital computer system market, specifically computers designed primarily for business, and subsequently alleged that IBM violated the antitrust laws in IBM's actions directed against leasing companies and plug-compatible peripheral manufacturers. Shortly after, IBM unbundled its software and services in what many observers believed was a direct result of the lawsuit, creating a competitive market for software. In 1982, the Department of Justice dropped the case as "without merit".[35]

Also in 1969, IBM engineer Forrest Parry invented the magnetic stripe card that would become ubiquitous for credit/debit/ATM cards, driver's licenses, rapid transit cards and a multitude of other identity and access control applications. IBM pioneered the manufacture of these cards, and for most of the 1970s, the data processing systems and software for such applications ran exclusively on IBM computers. In 1974, IBM engineer George J. Laurer developed the Universal Product Code.[36] IBM and the World Bank first introduced financial swaps to the public in 1981, when they entered into a swap agreement.[37]

The IBM Personal Computer *(pictured)* became one of IBM's best selling products and has had a wide influence on personal computing since its release in 1981.

IBM entered the microcomputer market in the 1980s with the IBM Personal Computer (IBM 5150). The computer, which spawned a long line of successors, had a profound influence on the development of the personal computer market and became one of

IBM's best selling products of all time. Because of a lack of foresight by IBM,[38][39] the PC was not well protected by intellectual property laws. As a consequence, IBM quickly began losing its market dominance to emerging, compatible competitors in the PC market.

In 1985, IBM collaborated with Microsoft to develop a new operating system, which was released as OS/2. Following a dispute, Microsoft severed the collaboration and IBM continued development of OS/2 on its own but it failed in the marketplace against Microsoft's Windows during the mid-1990s.

1990s–2000s

In 1991 IBM began spinning off its many divisions into autonomous subsidiaries (so-called "Baby Blues") in an attempt to make the company more manageable and to streamline IBM by having other investors finance those companies.[40][41] These included AdStar, dedicated to disk drives and other data storage products; IBM Application Business Systems, dedicated to mid-range computers; IBM Enterprise Systems, dedicated to mainframes; Pennant Systems, dedicated to mid-range and large printers; Lexmark, dedicated to small printers; and more.[42] Lexmark was acquired by Clayton & Dubilier in a leveraged buyout shortly after its formation.[43]

In 1997 Deep Blue, a chess computer created by IBM beat Chess World Champion at the time, Garry Kasparov in a six round match following standard tournament rules.

In September 1992, IBM finished spinning off its various microcomputer personal computer manufacturing divisions and were consolidated into an autonomous wholly owned subsidiary known as the IBM Personal Computer Company (IBM PC Co.).[44][45] This corporate restructuring came after IBM reported a sharp drop in profit margins during the second quarter of fiscal year 1992; market analysts attributed the drop to a fierce price war in the personal computer market over the summer of 1992.[46] The corporate restructuring was one of the largest and most expensive in history up to that point.[47] By the summer of 1993, the IBM PC Co. had divided into multiple business units itself, including Ambra Computer Corporation and the IBM Power Personal Systems Group, the former an attempt to design and market "clone" computers of IBM's own architecture and the latter responsible for IBM's PowerPC-based workstations.[48][49] IBM PC Co. introduced the ThinkPad clone computers, which IBM would heavily market and would eventually become one of the best-selling series of notebook computers.[50]

In 1993, IBM posted an $8 billion loss – at the time the biggest in American corporate history.[51] Lou Gerstner was hired as CEO from RJR Nabisco to turn the company around.[52] In 1995, IBM purchased Lotus Software, best known for its Lotus 1-2-3 spreadsheet software.[53] During the decade, IBM was working on a new operating

system, named the Workplace OS project. Despite a large amount of money spent on the project, it was cancelled in 1996.

IBM inventions (clockwise from top-left): the hard-disk drive, DRAM, the UPC bar code, and the magnetic stripe card

In 1998, IBM merged the enterprise-oriented Personal Systems Group of the IBM PC Co. into IBM's own Global Services personal computer consulting and customer service division. The resulting merged business units then became known simply as IBM Personal Systems Group.[54] A year later, IBM stopped selling their computers at retail outlets after their market share in this sector had fallen considerably behind competitors Compaq and Dell.[55] Immediately afterwards, the IBM PC Co. was dissolved and merged into IBM Personal Systems Group.[56]

In the 2000s, IBM collaborated with the Chinese defense contractor Huadi to develop the “Golden Shield Project”, a large-scale policing and surveillance project in Beijing. The system was reportedly used to monitor online activity and target groups such as Falun Gong, Uyghurs and other individuals considered security risks.[57][58][59]

In 2002 IBM acquired PwC Consulting, the consulting arm of PwC which was merged into its IBM Global Services.[60][61] On September 14, 2004, LG and IBM announced that their business alliance in the South Korean market would end at the end of that year. Both companies stated that it was unrelated to the charges of bribery earlier that year.[62][63][64][65] Xnote was originally part of the joint venture and was sold by LG in 2012.[66]

Continuing a trend started in the 1990s of downsizing its operations and divesting from commodity production, IBM sold all of its personal computer business to Chinese technology company Lenovo,[67] and in 2009, it acquired software company SPSS Inc. Later in 2009, IBM's Blue Gene supercomputing program was awarded the National Medal of Technology and Innovation by U.S. President Barack Obama.

2010s–present

In 2011, IBM gained worldwide attention for its artificial intelligence program Watson, which was exhibited on *Jeopardy!* where it won against game-show champions Ken Jennings and Brad Rutter. The company also celebrated its 100th anniversary in the same year on June 16. In 2012, IBM announced it had agreed to buy Kenexa and Texas Memory Systems,[68] and a year later it also acquired SoftLayer Technologies, a web hosting service, in a deal worth around $2 billion.[69] Also that year, the company designed a video surveillance system for Davao City.[70]

In 2014 IBM announced it would sell its x86 server division to Lenovo for $2.1 billion[71] while continuing to offer Power ISA-based servers.[72] Also that year, IBM began announcing several major partnerships with other companies, including Apple Inc.,[73][74] Twitter,[75] Facebook,[76] Tencent,[77] Cisco,[78] UnderArmour,[79] Box,[80] Microsoft,[81] VMware,[82] CSC,[83] Macy's,[84] Sesame Workshop,[85] the parent company of Sesame Street, and Salesforce.com.[86]

In 2015, its chip division transitioned to a fabless model with semiconductors design, offloading manufacturing to GlobalFoundries.[87]

In 2015, IBM announced three major acquisitions: Merge Healthcare for $1 billion,[88] data storage vendor Cleversafe, and all digital assets from The Weather Company,[89] including Weather.com and The Weather Channel mobile app.[90][91] Also that year, IBM employees created the film *A Boy and His Atom*, which was the first molecule movie to tell a story. In 2016, IBM acquired video conferencing service Ustream and formed a new cloud video unit.[92][93] In April 2016, it posted a 14-year low in quarterly sales.[94] The following month, Groupon sued IBM accusing it of patent infringement, two months after IBM accused Groupon of patent infringement in a separate lawsuit.[95]

In 2016, IBM bought Truven Health Analytics for $2.6 billion. In October 2018, IBM announced its intention to acquire Red Hat for $34 billion,[96][97][98] which was completed on July 9, 2019.[99]

In February 2020, IBM's John Kelly III joined Brad Smith of Microsoft to sign a pledge with the Vatican to ensure the ethical use and practice of Artificial Intelligence (AI).[100]

IBM announced in October 2020 that it would divest the Managed Infrastructure Services unit of its Global Technology Services division into a new public company.[101] The new company, Kyndryl, will have 90,000 employees, 4,600 clients in 115 countries, with an order backlog of $60 billion.[102][103][104] IBM's spin-off was greater than any of its previous divestitures, and welcomed by investors.[105][106][107] IBM appointed Martin

Schroeter, who had been IBM's CFO from 2014 through the end of 2017, as CEO of Kyndryl.[108][109]

In 2021, IBM announced the acquisition of the enterprise software company Turbonomic for $1.5 billion.[110] In January 2022, IBM announced it would sell Watson Health to private equity firm Francisco Partners.[111]

On March 7, 2022, a few days after the start of the Russian invasion of Ukraine, IBM CEO Arvind Krishna published a Ukrainian flag and announced that "we have suspended all business in Russia". All Russian articles were also removed from the IBM website.[112] On June 7, Krishna announced that IBM would carry out an "orderly wind-down" of its operations in Russia.[113]

In late 2022, IBM started a collaboration with new Japanese manufacturer Rapidus,[114] which led GlobalFoundries to file a lawsuit against IBM the following year.[115]

In 2023, IBM acquired Manta Software Inc. to complement its data and A.I. governance capabilities for an undisclosed amount.[116] On November 16, 2023, IBM suspended ads on Twitter after ads were found next to pro-Nazi content.[117][118]

In August 2023, IBM agreed to sell The Weather Company to Francisco Partners for an undisclosed sum.[119] The sale was finalized on February 1, 2024,[120] and the price was disclosed as $1.1 billion, with $750 million in cash, $100 million deferred over seven years, and $250 million in contingent consideration.[121]

In December 2023, IBM announced it would acquire Software AG's StreamSets and webMethods platforms for €2.13 billion ($2.33 billion).[122]

In June 2025, IBM was named by a UN expert report as one of several companies "central to Israel's surveillance apparatus and the ongoing Gaza destruction."[123][124]

On December 8, 2025, IBM announced a deal to acquire data-infrastructure company Confluent for approximately $11 billion. The deal is expected to help further advance the company's interests in AI.[125]